# HE2003 Econometrics I
# Tutorial 8

21/10/2024

# Question 1

a) Give an example of linear regression model with single regressor (X1) and omitted variable (X2) that has concern of OVB.

Example: $X_1$ is student-teacher ratio (STR) while $X_2$ is a certain demographic factor (i.e. household income) that both affects the student performance ($Y$) and correlates with STR.

In general, as long as we can show that $X_1$ (included regressor) and $X_2$ (omitted variable) are correlated, and $X_2$ has effect on $Y$, then we can say that there exists concern of OVB from $X_2$ (in practice, how serious is this concern will depend on whether $X_1$ and $X_2$ are highly correlated and also the value of the coefficient $\beta_2$). Also, the sign of the OVB can be seen from the following table:

| | $Cov(X_1, X_2) > 0$ | $Cov(X_1, X_2) < 0$ |
|---|---|---|
| $\beta_2 > 0$ | Positive (upward) bias | Negative (downward) bias |
| $\beta_2 < 0$ | Negative (downward) bias | Positive (upward) bias |

# Question 1

b) Explain the meaning of dummy variable trap (DVT). Give an example of DVT.

The multicollinearity arises when one of the regressors is a perfect linear combination of the other regressors. An example of multicollinearity is the dummy variable trap, which occurs when we use $G$ dummy variables for $G$ categories (when we already have a constant term in the model).

For instance, there are 3 possible areas: suburb, urban and rural. Define $Suburb_i = 1$ if the $i$-th district is in suburb area and 0 otherwise, $Urban_i = 1$ if the $i$-th district is in urban area and 0 otherwise, and $Rural_i = 1$ if the $i$-th district is in rural area and 0 otherwise.

Then, the following model has the **dummy variable trap**:

$$Testscore_i = \beta_0 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i \,, \qquad i = 1, \ldots, n$$

# Question 1

The model can be written as:

$$Testscore_i = \beta_0 \times 1 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i\,, \qquad i = 1, \dots, n$$

For all $i$ in the data:

$$Suburb_i + Urban_i + Rural_i = 1$$

Here, 1 corresponds to the constant term [the implicit regressor for the intercept; therefore, here we in fact have 4 explicit regressors ($STR$, $Suburb$, $Urban$, and $Rural$) and 1 implicit regressor].

This problem can be solved easily by **omitting any one of the three dummy variables**. For example, if we omit $Suburb_i$, then we will have $Urban_i + Rural_i \neq 1$ for some $i$.

We cannot solve the dummy variable trap by simply changing 1 to some other number such as 2:

$$Testscore_i = \beta_0 \times 2 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i\,, \qquad i = 1, \dots, n$$

The dummy variable trap still exists because for all $i$ in the data:

$$2(Suburb_i + Urban_i + Rural_i) = 2$$

# Question 1

Similarly, for the gender-wage-gap model discussed in class, suppose we have the model:

$$Wage_i = \beta_0 \times 1 + \beta_1 D_{1,i} + \beta_2 D_{2,i} + u_1 \,, \qquad i = 1, \dots, n$$

where we define:

- $D_{1,i} = 1$ if $i$ is male and 0 if $i$ is female
- $D_{2,i} = 1$ if $i$ is female and 0 if $i$ is male

This model is not estimable because of the dummy variable trap because for all $i$:

$$D_{1,i} + D_{2,i} = 1$$

(assume there are only 2 genders in this case)

# Question 2

The following equation describes the housing price in a community in terms of amount of pollution ($nox$ for nitrous oxide) and the number of rooms in houses in the community ($rooms$):

$$price_i = \beta_0 + \beta_1 nox_i + \beta_2 rooms_i + e_i, \qquad i = 1, \ldots, n;$$

$$E[e_i | nox_i, rooms_i] = 0$$

(a) Explain the two conditions of omitted variable bias.

(b) What are the probable signs of $\beta_1$ and $\beta_2$?

(c) Suppose that $nox_i$and $rooms_i$ are negatively correlated. Does the simple regression of $price_i$ on $nox_i$ produce an upward or a downward biased estimator of $\beta_1$? Explain.

# Question 2

a)

The OVB is the bias in the OLS estimator that arises when the explanatory variable is correlated with a variable that is omitted from the regression. The OVB occurs if

the following two conditions hold:

(1) The explanatory variable (regressor) is correlated with the omitted variable;

(2) The omitted variable is a determinant of the dependent variable (the omitted variable has an effect on the dependent variable).

b)

The probable signs of coefficients: $\beta_1$ negative and $\beta_2$ positive.

Pollution should have a negative impact on housing price while the number of rooms is positively correlated with price.

# Question 2

c)

The simple regression model with $rooms_i$ omitted can be written as:

$$price_i = \beta_0 + \beta_1 nox_i + u_i, \qquad i = 1, \dots, n;$$

where $u_i = \beta_2 rooms_i + e_i$.

From the OVB formula:

$$OVB = \beta_2 \frac{Cov(nox_i, rooms_i)}{Var(nox_i)}$$

Note that $\beta_2$ is positive, and $Cov(nox_i, rooms_i)$ is negative because $Corr(nox_i, rooms_i)$ is negative. Therefore, the OVB is negative, which means that the simple regression of $price_i$ on $nox_i$ produces a downward biased estimator of $\beta_1$.

# Question 3

This question is based on the 2015 Current Population Survey in US. The dataset consists of information on 7178 full-time, full-year workers. The highest educational achievement for each worker was either a high school diploma or a bachelor's degree. The workers' ages ranged from 25 to 34 years. The dataset also contains information on the region of the country where the person lived, marital status, and number of children.

Let $AHE$ = average hourly earnings,

$College$ = binary variable (1 if college, 0 if high school),

$Female$ = binary variable (1 if female, 0 if male),

$Age = age$ (in years),

$Northeast$ = binary variable (1 if $Region = Northeast$, 0 otherwise), $Midwest$ = binary variable (1 if $Region = Midwest$, 0 otherwise), $South$ = binary variable (1 if $Region = South$, 0 otherwise), $West$ = binary variable (1 if $Region = West$, 0 otherwise).

# Question 3

**Results of Regressions of Average Hourly Earnings on Sex and Education Binary Variables and Other Characteristics, Using 2015 Data from the Current Population Survey**

**Dependent variable: average hourly earnings (*AHE*).**

| Regressor | (1) | (2) | (3) |
|---|---|---|---|
| College ($X_1$) | 10.47 | 10.44 | 10.42 |
| Female ($X_2$) | −4.69 | −4.56 | −4.57 |
| Age ($X_3$) | | 0.61 | 0.61 |
| Northeast ($X_4$) | | | 0.74 |
| Midwest ($X_5$) | | | −1.54 |
| South ($X_6$) | | | −0.44 |
| Intercept | 18.15 | 0.11 | 0.33 |

# Question 3

a) Using the regression results in column (1): Do workers with college degrees earn more, on average, than workers with only high school diplomas? How much more?

The multiple regression model in column (1) can be written as:

$$AHE_i = \beta_0 + \beta_1 College_i + \beta_2 Female_i + u_i\ , \qquad i = 1, \ldots, n$$

Then, according to the results in column (1), on average, workers with college degrees earn more, than workers with only high school diplomas by 10.47 (dollars per hour).

# Question 3

b) Using the regression results in column (2): Sally is a 29-year-old female college graduate. Besty is a 34-year-old female college graduate. Predict Sally's and Besty's earnings.

The multiple regression model in column (1) can be written as:

$$AHE_i = \beta_0 + \beta_1 College_i + \beta_2 Female_i + \beta_3 Age_i + u_i\,, \qquad i = 1, \dots, n$$

Then, according to the results in column (2), the prediction for Sally is:

$$\widehat{AHE} = 0.11 + 10.44 \times 1 - 4.56 \times 1 + 0.61 \times 29 = 23.68 \text{ (dollars per hour)}$$

The prediction for Besty is:

$$\widehat{AHE} = 0.11 + 10.44 \times 1 - 4.56 \times 1 + 0.61 \times 34 = 26.73 \text{ (dollars per hour)}$$

# Question 3

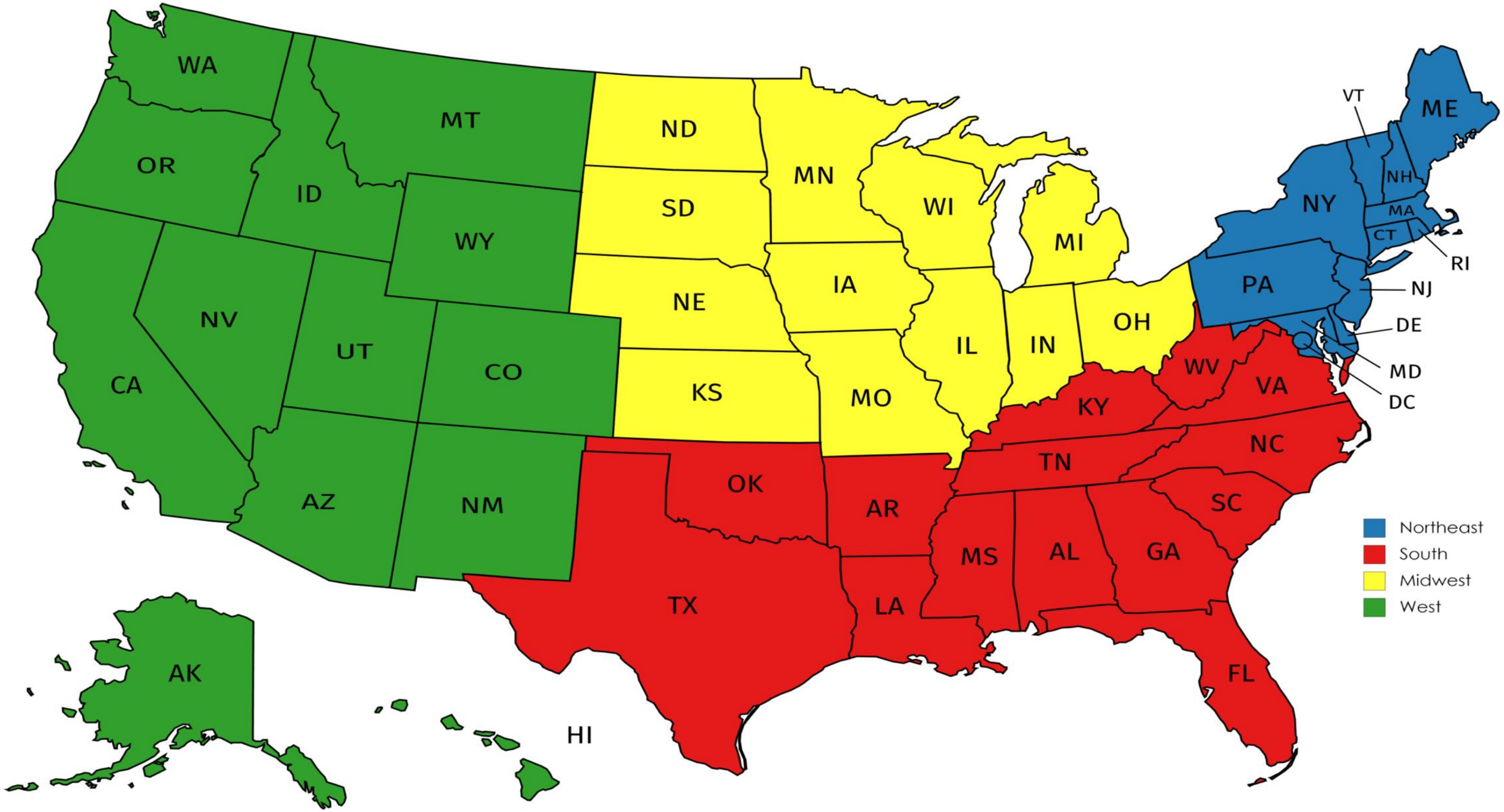
c) Using the regression results in column (3):

   i. Why is the regressor $West$ omitted from the regression? What would happen if it were included?

   ii. Juanita is a 28-year-old female college graduate from the South. Jennifer is a 25-year-old female college graduate from the Midwest. Calculate the predicted difference in earnings between the two.

$West$ is omitted from the regression to avoid the dummy variable trap. As discussed in question 1(b), if it were included there would be a perfect linear combination among the regressors (that is, $Northeast, Midwest, South, West$ and the constant term)

# Question 3


WA
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GA
FL
SC
NC
VA
WV
PA
NY
VT
NH
ME
MA
CT
RI
NJ
DE
MD
DC
Northeast
South
Midwest
West

# Question 3

The multiple regression model in column (3) can be written as:

$$AHE_i = \beta_0 + \beta_1 College_i + \beta_2 Female_i + \beta_3 Age_i + \beta_4 Northeast_i$$
$$+\beta_5 Midwest_i + \beta_6 South_i + u_i\,, \qquad i = 1, \dots, n$$

Then, the prediction for Juanita is:

$$\widehat{AHE} = 0.33 + 10.42 \times 1 - 4.57 \times 1 + 0.61 \times 28 + 0.74 \times 0$$
$$-1.54 \times 0 - 0.44 \times 1 = 22.82 \text{ (dollars per hour)}$$

The prediction for Jennifer is:

$$\widehat{AHE} = 0.33 + 10.42 \times 1 - 4.57 \times 1 + 0.61 \times 25 + 0.74 \times 0$$
$$-1.54 \times 1 - 0.44 \times 0 = 19.89 \text{ (dollars per hour)}$$

Therefore, the predicted difference in earnings between the two is equal to $22.82 - 19.89 = 2.93$ (dollars per hour, which is also equal to $0.63 \times 3 - 0.44 + 1.54$).

# Question 4

Empirical exercise: Use the dataset **birthweight_smoking.dta** uploaded in NTULearn to answer the following questions (STATA is available in the computer labs at B1 of the SHHK building, but you are welcome to use other software if you prefer).

(a) Regress “birthweight” (birth weight of infant, in grams) on “smoker” (indicator variable equal to one if the mother smoked during pregnancy, and zero otherwise). What is the estimated effect of smoking on birth weight?

(b) Regress “birthweight” on “smoker” and “nprevisit” (total number of prenatal care visits). How is the estimated effect of smoking different from the result in (a)? Show the sign of the potential omitted variable bias of the regression with the single regressor “smoker”. (Note: In STATA, we can check the correlation between two variables X and Y using the following command: **correlate X Y**)

(c) Jane smoked during her pregnancy and had 8 prenatal care visits. Use the regression to predict the birth weight of Jane’s child.

# Question 4

```
. regress birthweight smoker, robust

Linear regression                               Number of obs     =      3,000
                                                F(1, 2998)        =      89.21
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0286
                                                Root MSE          =     583.73

------------------------------------------------------------------------------
             |               Robust
 birthweight |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      smoker |  -253.2284   26.81039    -9.45   0.000     -305.797   -200.6597
       _cons |    3432.06   11.89053   288.64   0.000     3408.746    3455.374
------------------------------------------------------------------------------
```

With the simple regression, the estimated effect of smoking on birth weight is -253.23 grams

# Question 4

```
. reg birthweight smoker nprevist, robust

Linear regression                               Number of obs     =      3,000
                                                F(2, 2997)        =      89.16
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0728
                                                Root MSE          =     570.39
```

| birthweight | Coef. | Robust Std. Err. | t | P>\|t\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| smoker | -218.8294 | 25.98922 | -8.42 | 0.000 | -269.7879 | -167.8709 |
| nprevist | 34.10394 | 3.608863 | 9.45 | 0.000 | 27.02784 | 41.18004 |
| _cons | 3050.527 | 43.69276 | 69.82 | 0.000 | 2964.857 | 3136.198 |

# Question 4

With the multiple regression, the estimated effect of smoking on birth weight is -218.83 grams. The difference between the simple regression and multiple regression is 34.4 grams, or around 16% of the estimated result (34.4/218.83 ≈ 0.16).

The multiple regression model is:

$$birthweight_i = \beta_0 + \beta_1 smoker_i + \beta_2 nprevist_i + u_i\ , \qquad i = 1, \dots, n$$

To discuss the OVB, we first need to determine the signs of $\beta_2$ and $Cov(smoker_i, nprevist_i)$.

```
. correlate smoker nprevist
(obs=3,000)
```

| | smoker | nprevist |
|---|---|---|
| smoker | **1.0000** | |
| nprevist | **-0.1086** | **1.0000** |

# Question 4

It is reasonable to think that number of prenatal visit will have positive effect on baby's birth weight, so $\beta_2$ is positive. In addition, the correlation between "smoker" and "nprevist" is -0.11 so their covariance should also be negative.

$$OVB = \beta_2 \frac{Cov(smoker_i, nprevist_i)}{Var(smoker_i)}$$

Therefore, we conclude that the OVB is negative by using the OVB formula, which means that the simple regression in (a) produces downward bias, and this conclusion is in fact coherent with the actual estimation result (-253.23 vs -218.83).

A natural explanation of the negative correlation between "smoker" and "nprevist" is that if a mother smokes during pregnancy, then she may be relatively careless for the new baby, so she is less likely to go to hospitals or clinics for prenatal care visits.

# Question 4

$$\widehat{birthweight} = \hat{\beta}_0 + \hat{\beta}_1 smoker + \hat{\beta}_2 nprevist = 3050.53 - 218.83 \times 1 + 34.1 \times 8 = 3104.5\ (grams)$$

```
. margins, at(smoker=1 nprevist=8)

Adjusted predictions                            Number of obs     =      3,000
Model VCE    : Robust

Expression   : Linear prediction, predict()
at           : smoker          =           1
               nprevist        =           8
```

| | Margin | Delta-method Std. Err. | t | P>\|t\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| _cons | 3104.53 | 25.26846 | 122.86 | 0.000 | 3054.984 | 3154.075 |

# Question 4

```
. summarize
```

| Variable | Obs | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| nprevist | 3,000 | 10.99167 | 3.672069 | 0 | 35 |
| birthweight | 3,000 | 3382.934 | 592.1629 | 425 | 5755 |
| smoker | 3,000 | .194 | .3954948 | 0 | 1 |

```
. table smoker
```

| smoker | Freq. |
|---|---|
| 0 | 2,418 |
| 1 | 582 |